QUESTION 1

What type of programming language is typically used in Arduino sketches?

Answer is Correct **Score: 1 / 1**

Options

A. Pure C

B. Assembly

Correct

C. C variant specific to Arduino

D. Python

FEEDBACK

Solution: Arduino uses a variant of C with its own conventions, not pure C or C++.

Link: https://youtu.be/D8-wnzT6MQQ?feature=shared&t=27

QUESTION 2

What are the two essential functions in every Arduino sketch?

Answer is Correct **Score: 1 / 1**

Options

A. init and loop

Correct

B. setup and loop

C. main and setup

D. run and halt

FEEDBACK

Solution: Every Arduino sketch must have setup() and loop() functions which the system uses internally.

Link: https://youtu.be/D8-wnzT6MQQ?feature=shared&t=847

QUESTION 3

What is the purpose of the #define LED D13 directive in Arduino code?

Answer is Wrong **Score: 0 / 1**

Options

A. Declares an LED array

Expected

B. Maps the LED pin to D13 for readability

C. Defines a digital input

Wrong

D. Compiles a macro into machine code

FEEDBACK

Solution: The #define macro replaces LED with D13 to improve code readability

Link: https://youtu.be/D8-wnzT6MQQ?feature=shared&t=702

QUESTION 4

Why is pinMode(LED, OUTPUT) used in the setup() function?

Answer is Correct **Score: 1 / 1**

Options

A. To read from the LED pin

B. To define the LED pin as input

Correct

C. To set the LED pin as output

D. To turn on the LED

FEEDBACK

Solution: pinMode() configures the direction of the pin; here, it's set as an output.

Link : https://youtu.be/D8-wnzT6MQQ?feature=shared&t=905

QUESTION 5

Which function is responsible for sending HIGH or LOW signals in Arduino?

Answer is Correct **Score: 1 / 1**

Options

A. analogWrite()

B. pinRead()

C. digitalRead()

Correct

D. digitalWrite()

FEEDBACK

Solution: digitalWrite(pin, value) sends HIGH or LOW signals to a digital output pin.

Link: https://youtu.be/D8-wnzT6MQQ?feature=shared&t=976

QUESTION 6

What does delay(500) mean in Arduino code?

Answer is Wrong **Score: 0 / 1**

Options

A. Delay for 5 seconds

Wrong

B. Delay for 500 microseconds

Expected

C. Delay for 0.5 seconds

D. Wait until input is received

FEEDBACK

Solution: delay(500) introduces a 500-millisecond delay, i.e., half a second.

Link: https://youtu.be/D8-wnzT6MQQ?feature=shared&t=993

QUESTION 7

What is the advantage of using #define for pin names in Arduino code?

Answer is Correct **Score: 1 / 1**

Options

A. Optimizes memory usage

B. Increases voltage

Correct

C. Makes code more readable and abstract

D. Reduces compilation time

FEEDBACK

Solution: Macros like #define LED D13 abstract away raw pin numbers, making code easier to understand.

Link: https://youtu.be/D8-wnzT6MQQ?feature=shared&t=726

QUESTION 8

What happens in loop() function of an Arduino sketch?

Answer is Correct **Score: 1 / 1**

Options

A. It gets called once

B. It handles setup configurations

Correct

C. It gets executed repeatedly

D. It initializes hardware

FEEDBACK

Solution: After setup(), the loop() function runs endlessly unless power is removed.

Link: https://youtu.be/D8-wnzT6MQQ?feature=shared&t=859

QUESTION 9

Why does the LED blink with a 1-second period in the sketch?

Answer is Correct **Score: 1 / 1**

Options

A. Due to 1ms delay

Correct

B. Because delay(500) is used twice

C. Pin is pulsed externally

D. 1 Hz timer is configured

FEEDBACK

Solution: The LED is turned on for 500 ms and off for 500 ms, totaling a 1-second period.

Link: https://youtu.be/D8-wnzT6MQQ?feature=shared&t=998

QUESTION 10

Why might a GPIO pin not toggle even with correct code?

Answer is Correct **Score: 1 / 1**

Options

A. LED is not connected

B. Board is unpowered

Correct

C. Pin is not configured as output

D. Pin number is odd

FEEDBACK

Solution: GPIO pins are inputs by default and must be explicitly configured as outputs to control an LED.

Link: https://youtu.be/D8-wnzT6MQQ?feature=shared&t=2091

QUESTION 11

What is the role of the digitalWrite(LED, HIGH) in Arduino sketches?

Answer is Correct **Score: 1 / 1**

Options

A. Reads the LED state

Correct

B. Turns the LED on

C. Resets the microcontroller

D. Compiles the sketch

FEEDBACK

Solution: It sends a HIGH signal to the pin connected to the LED, turning it on.

Link: https://youtu.be/D8-wnzT6MQQ?feature=shared&t=981

QUESTION 12

What can be abstracted away using Arduino functions like pinMode and digitalWrite?

Answer is Correct **Score: 1 / 1**

Options

A. Memory map layout

Correct

B. Register configuration

C. Clock setup

D. Analog pin resolution

FEEDBACK

Solution: Arduino functions abstract away the register-level operations, simplifying hardware access.

Link: https://youtu.be/D8-wnzT6MQQ?feature=shared&t=750